

$$v = \sqrt{\frac{k}{m}}$$

$$T = \frac{2\pi}{\sqrt{\frac{k}{m}}}$$

$$T = \frac{2\pi}{\omega}$$

$$x = x_m \sin(\sqrt{\frac{k}{m}}t + \delta)$$

$$x = x_m \cos(\sqrt{\frac{k}{m}}t + \delta)$$

$$\sin \omega t$$

$$\frac{2\pi}{T}$$

$$\sin \frac{t}{2}$$

$$\frac{2\pi}{T} = \omega$$

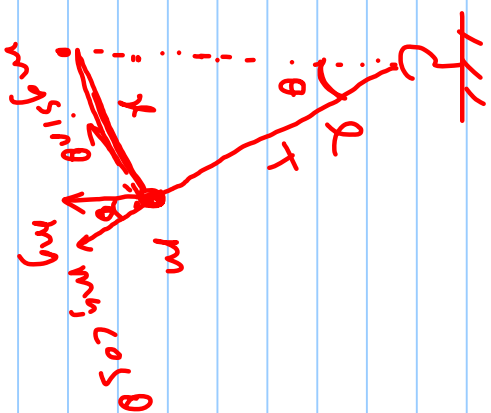
$$f(t) = f(t + T)$$

period

$$\sin t = \sin(t + 2\pi)$$

$$\downarrow 2\pi$$

Simple pendulum



periodic

ω

SHM

Restoring

$$F = -mg \sin \theta$$

Force

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

$$10^\circ \leq \theta \leq 40^\circ \quad \sin \theta \approx \theta$$

$$\sin \theta \approx \frac{x}{l}$$

$$K = \frac{mgl}{2}$$

$$F = -mgl\theta$$

$$F = -mgl\frac{x}{l}$$

$$F = -\frac{mgl}{l}x$$



$$F = -kx$$

$$\Rightarrow m \frac{d^2x}{dt^2} + \frac{mgl}{l}x = 0$$

$\Rightarrow$

$$\left[\frac{d^2x}{dt^2} + \frac{g}{l}x = 0 \right]$$

$$\omega^2 = \frac{g}{l}$$

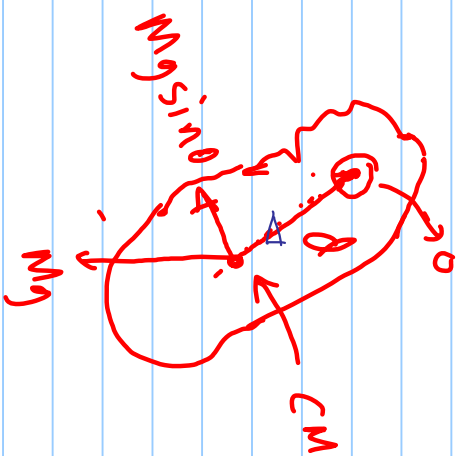
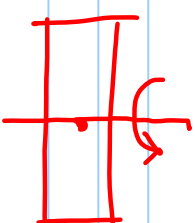
$$\omega = \sqrt{\frac{g}{l}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}$$

$$x = x_m \sin(\omega t + \phi)$$

V
 a

Physical Pendulum,



$$F = -kx$$

$$\tau = -k\theta$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = -d Mg \sin \theta$$

$$\tau = -Mgd \sin \theta$$

$$\sin \theta \approx \theta$$

$$\tau = -mgl \theta \quad k = mgl$$

$$\Rightarrow I \ddot{\theta} = -mgl \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} + \frac{mgl}{I} \theta = 0 \quad \omega$$

$$\omega = \sqrt{\frac{mgl}{I}}$$

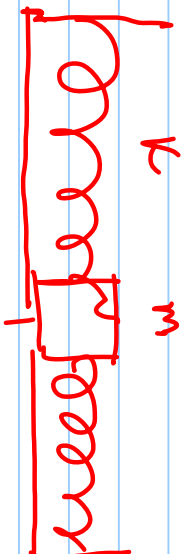
$$T = 2\pi \sqrt{\frac{I}{mgl}}$$

$$T = \frac{2\pi}{\sqrt{\frac{mgl}{I}}}$$

$$T = 2\pi \sqrt{\frac{I}{k}}$$

$$\begin{cases} \theta = \theta_m \sin(\omega t + \delta) \\ \frac{d\theta}{dt} \end{cases}$$

$$m a = -k x$$



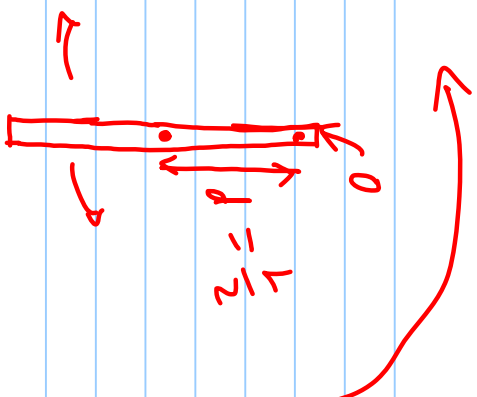
$$\frac{d^2 x}{dt^2} + \left(\frac{2k}{m} \right) x$$

$$m a = -k x - k x$$

$$\Rightarrow m \frac{d^2 x}{dt^2} + 2k x = 0 \quad \omega = \sqrt{\frac{2k}{m}}$$

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$I = \frac{1}{3} ML^2$$

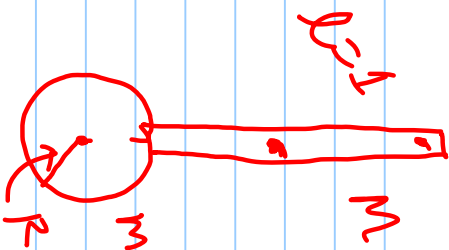


$$L = 1\text{ m}$$

$$M = 4\text{ kg}$$

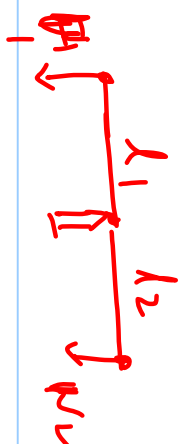
$$M = \frac{m_1 + m_2}{m_1 + m_2}$$

$$d = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$



$$T = 2\pi \sqrt{\frac{I}{Mgd}}$$

$$I = \frac{1}{3} ML^2 + m(L+R)^2 + \frac{1}{2} mR^2$$



$$w_1 l_1 = w_2 l_2$$